by i), taking into account the randomness in the decoding process (e.g., probabilistic sampling of the next token). Finally, the two-dimensional loss landscape function centered on the input embedding $\mathbf{E}(\mathbf{x})$ is computed by $F(\alpha, \beta|\mathbf{E}(\mathbf{x})) = f(\mathbf{E}(\mathbf{x}) \oplus \alpha \cdot \mathbf{v}_1 \oplus \beta \cdot \mathbf{v}_2|g, J)$, where $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^d$ are two random directions sampled from a standard multivariate Gaussian distribution, and the notation $\oplus$ denotes the row-wise broadcast function that adds a vector to each row of $\mathbf{E}(\mathbf{x})$.

Figure 2 shows the average loss landscapes of benign and jailbreak prompts evaluated with Vicuna-7B-V1.5 (Zheng et al., 2024), a fine-tuned LLM based on Meta's Llama-2-7B model (Touvron et al., 2023). The benign prompts are 100 queries sampled from Chatbot Arena[1], while the 100 jailbreak prompts are generated by the Greedy Coordinate Gradient (GCG) attack on AdvBench for suffix optimization (Zou et al., 2023). Keyword matching is used as the judge function (Hu et al., 2024). The results show that benign and malicious prompts differ significantly in their respective loss landscapes. The loss landscape of benign prompts is flat and close to 1, suggesting that the perturbed input embeddings are unlikely to cause rejections. On the other hand, malicious prompts are sensitive to embedding perturbations. The average non-refusal rate peaks at the origin (because most jailbreak attempts are successful), but quickly declines as the input embeddings are perturbed away from the origin, suggesting that the perturbed embeddings of malicious prompts are more likely to be refused by the LLM. Based on the loss landscape analysis, Hu et al. propose a jailbreak prompt detection method called Gradient Cuff (Hu et al., 2024). The detection of Gradient Cuff consists of two steps. In the first step, the detector flags a jailbreak if the non-refusal loss f is less than 0.5, which checks the tendency of the LLM to reject a query. In the second step, the detector computes the gradient norm of f at the input embedding $\mathbf{E}(\mathbf{x})$ and uses it to quantify the changes in the loss landscape under random perturbations. If the gradient norm is greater than a certain threshold, the query is predicted to be a jailbreak prompt. In practice, such a threshold can be determined by controlling the FPR evaluated on a validation dataset of benign prompts. In addition, if the judge function J is not differentiable, such as in the case of keyword matching, Gradient Cuff uses zeroth-order optimization (Liu et al., 2020), another signal processing technique for gradient estimation using only function evaluations, to approximate the gradient norm.

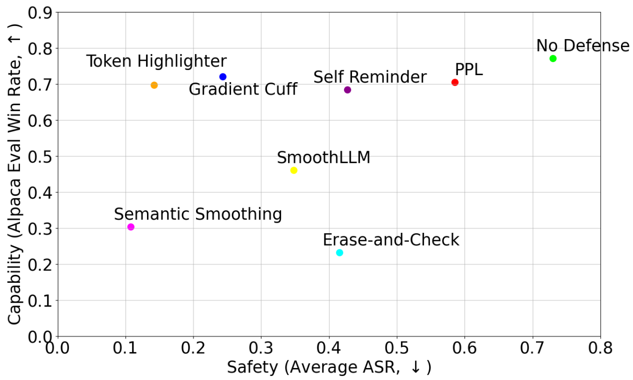
Mitigating Problematic Tokens via Sensitivity Analysis Beyond jailbreak prompt detection, Hu et al. explore a jailbreak mitigation framework called Token Highlighter (Hu et al., 2025) to identify and attenuate problematic tokens that could lead to jailbreaks. In particular, based on the principle that successful jailbreak prompts trick the target LLM into starting with an affirmative answer like "Sure, here's..." (Zou et al., 2023), Token Highlighter defines an affirmation loss as $\text{Loss}_{\text{Aff}} = -\log p_\theta(\mathbf{y}_a|\mathbf{x})$, where $\mathbf{x}$ is the context (query), $\mathbf{y}_a$ is an affirmation phrase, and $\log p_\theta(\cdot)$ is the log-likelihood computed by an LLM parameterized by θ. The phrase $\mathbf{y}_a$ = "Sure, I'd like to help you with this." is the default choice in (Hu et al., 2025). With the affirmation loss Loss_{Aff}, the influence of each token in $\mathbf{x}$ is measured by taking the gradient of Loss_{Aff} with respect to the input embedding matrix $\mathbf{E}(\mathbf{x}) := [\mathbf{e}_1; \mathbf{e}_2; \cdots \mathbf{e}_{|\mathbf{x}|}]^T$ and computing the gradient norm of each token, formally defined as $\{\|\nabla_{\mathbf{e}_j} \text{Loss}_{\text{Aff}}\|\}_{j=1}^{|\mathbf{x}|}$, where $\mathbf{e}_j \in \mathbb{R}^d$ is the token embedding of the j-th token in $\mathbf{x}$. Next, Token Highlighter takes the top $Q\%$ of tokens in $\mathbf{x}$ and proposes the concept of *soft removal* by shrinking the embeddings of these most influential tokens by γ, where $\gamma \in [0, 1)$. Finally, the LLM returns the response of the attenuated input embedding after soft removal as the ultimate model output. In its evaluation, the soft removal effectively mitigates jailbreak attempts for malicious queries while balancing the generation quality for benign queries.

Performance Evaluation We choose Vicuna-7B-V1.5 (Zheng et al., 2024) as the target LLM to study the safety-capability trade-offs of different jailbreak prompt mitigation methods. Vicuna-7B-V1.5 is an open-weight LLM with 7 billion parameters, with good capability and mediocre safety alignment, making it an ideal candidate for studying jailbreak attack and defense strategies. The following 6 jailbreak attacks are used for evaluation:

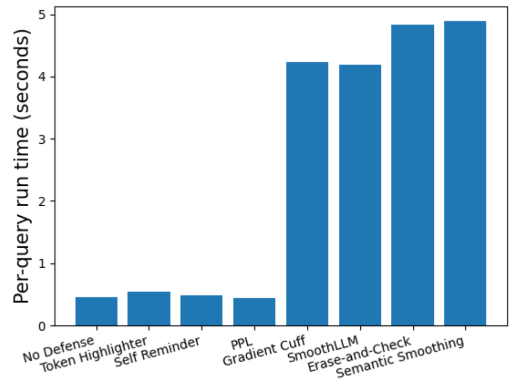
- GCG (Zou et al., 2023) trains and appends an adversarial suffix to a query using a greedy coordinate gradient (GCG) algorithm.
- AutoDAN (Liu et al., 2024) designs a hierarchical genetic algorithm for jailbreak prompt optimization.
- PAIR (Chao et al., 2025) uses LLMs as an attacker and a judge for prompt automatic iterative refinement (PAIR).
- TAP (Mehrotra et al., 2024) extends PAIR with reasoning ability based on Tree of Attacks with Pruning (TAP).
- AIM[2] is a role-playing jailbreak attack by asking the target LLM to be Always Intelligent and Machiavellian (AIM) to give an unfiltered response to any request.
- Many-shot (Anil et al., 2024) uses in-context examples for jailbreaking, where multiple fake dialogues between a user and an AI assistant (with affirmative responses) are provided as prefixes to the actual user query.

[1] https://lmarena.ai/

[2] The AIM prompt is obtained from https://github.com/alexalbertt/jailbreakchat



(a) Safety-capability analysis



(b) Run time analysis

Figure 3. Comparison of jailbreak prompt detection and mitigation methods. (a) Safety-capability trade-offs. The safety performance is evaluated by the attack success rate (ASR) averaged over 6 jailbreak attacks, and the capability performance is evaluated by the win rate in Alpaca Eval (Li et al., 2023). A higher win rate and lower ASR means a better approach. See the "Performance Evaluation" paragraph for details). (b) Per-query run time analysis (seconds). Overall, Token Highlighter is the most economical method that best balances the safety-capability trade-off with lightweight compuation cost.

We also consider the following 8 mitigation strategies in the evaluation.

- No Defense means the original LLM without any mitigation strategy.
- PPL (Jain et al., 2023) uses a perplexity (PPL) filter on the input query for detection.
- Erase-and-Check (Kumar et al., 2024) independently erases some tokens of the original query multiple times and checks if any of the remaining queries would be rejected by the target LLM.
- SmoothLLM (Robey et al., 2023) randomly perturbs multiple copies of the original query and then aggregates the corresponding predictions for detection.
- Semantic Smoothing (Ji et al., 2024) extends SmoothLLM with multiple semantically transformed copies.
- Self Reminder (Xie et al., 2023) changes the system prompt to remind the target LLM to respond responsibly.
- Gradient Cuff (Hu et al., 2024) explores the refusal loss landscape for detection.
- Token highlighter (Hu et al., 2025) uses the gradient of an affirmation loss and soft removal for mitigation.

For capability assessment, we report the *Win Rate* measured on Alpaca Eval (Li et al., 2023), where the protected LLM's response is compared with that of OpenAI's Text-Davinci-003 model, and the quality is judged by GPT-4. For safety evaluation, we sampled 100 harmful queries from AdvBench (Zou et al., 2023) as the prototype prompts to implement jailbreak attacks. We report the *Attack Success Rate* (ASR) of the input queries and their responses averaged over the aforementioned 6 jailbreak attacks, where the LLaMA-Guard-2 model (Inan et al., 2023) judges the success. An ideal mitigation strategy would have low ASR and high capability. We use the same implementations for the attacks and defenses as in (Hu et al., 2025).

Figure 3(a) shows the safety-capability trade-offs of the 8 mitigating strategies. Compared to the original (No Defense) LLM, it can be observed that Gradient Cuff and Token Highlighter strike a better trade-off between the win rate and ASR than other methods, achieving a significant reduction in the ASR while maintaining a similar level of capability. On the other hand, other methods, such as semantic smoothing, can further reduce the ASR at the cost of a more than 50% drop in the win rate, which could severely compromise the performance of the protected LLM. Figure 3(b) compares the per-query run time of each mitigation method. Mitigation strategies such as Gradient Cuff, SmoothLLM, Semantic Smoothing, and Erase-and-Check require additional inference on multiple modified copies of the original query, incurring high computation cost and latency. PPL and Self Reminder are cost-effective, but their defense performance is less effective than others. Overall, Token Highlighter is the most economical mitigation strategy, as it achieves a good tradeoff between safety and capability, and has lightweight computation.

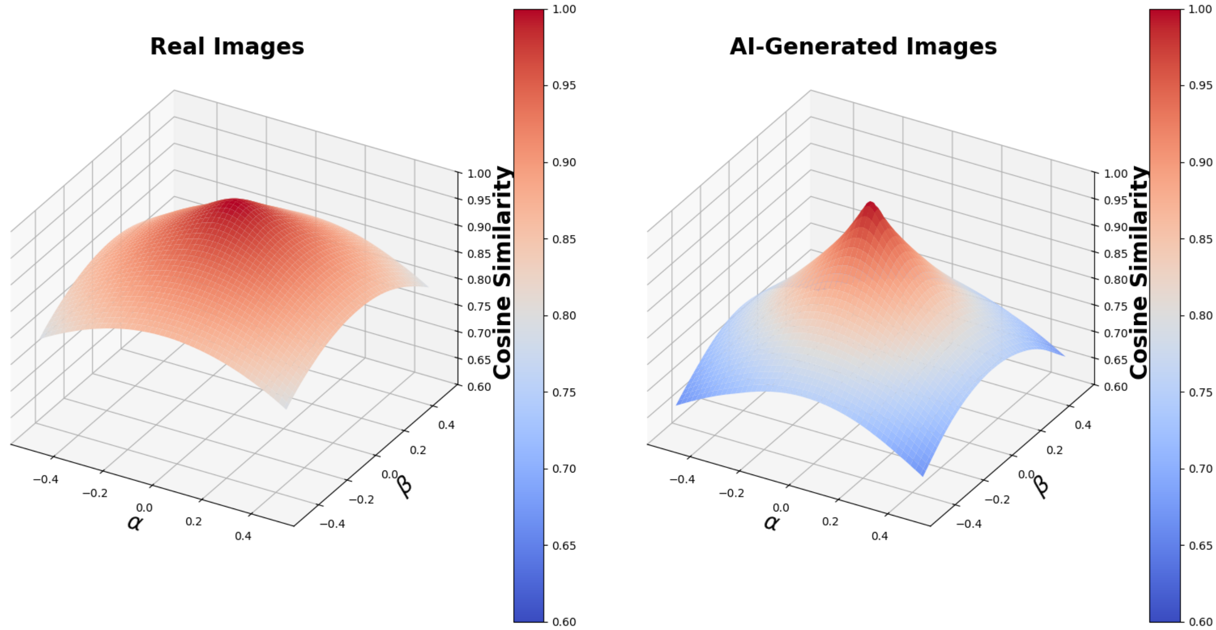


Figure 4. Loss landscape analysis for real and AI-generated images. We use the embedding of DINOV2 (Oquab et al., 2023) to compute the cosine similarity between an original image and its perturbed version by interpolating two random directions with additive Gaussian noise in pixel space, where the perturbation strengths are denoted by α and β. The results are averaged over 100 images. The real images are sampled from ImageNet, while the AI-generated ones are generated by the ablated diffusion mode (ADM) (Dhariwal & Nichol, 2021). The cosine similarity analysis shows that AI-generated images are more sensitive to Gaussian perturbations than real images.

5. Model Output Safety: AI-Generated Content Detection

With the rapidly increasing ability of GenAI tools to generate realistic, high-quality, and creative content across modalities, including but not limited to text, image, audio, and video, procedures to ensure that AI-generated content (AIGC) can be reliably validated are critical to the responsible use and governance of GenAI. Notable challenges associated with AIGC detection include misuse to create and disseminate disinformation at scale (e.g., AI-generated deepfakes), contamination of future training data, intellectual property protection, hallucinated responses, and inappropriate or unethical content generation. It is worth noting that watermarking is only a partial and somewhat limited solution to the challenge of AIGC detection, because watermarking only regulates responsible GenAI service providers, but not bad actors. A bad actor can train his own GenAI model or use any unregulated GenAI tool to create AIGC without watermarking. Therefore, watermarking only provides passive protection for AIGC, and reliable AIGC detectors without watermark assumption are in high demand as proactive AI safety tools. In this section, we discuss two AIGC detectors for image and text that are enhanced by signal processing techniques.

Table 2. AUROC scores of training-free AI-generated image detectors on three datasets: ImageNet, LSUN-Bedroom, and DF40 (deepfakes).

Method	ImageNet	LSUN-Bedroom	DF40 (Deepfake)
AEROBLADE	0.593	0.595	0.526
RIGID	0.867	0.877	0.735

AI-generated Image Detection via Sensitivity Analysis For the detection of AI-generated images, we focus on training-free approaches that are built upon off-the-shelf AI models, as recent work has shown comparable or even better detection performance of training-free methods than training-based approaches (He et al., 2024; Tsai et al., 2024). Recall that sensitivity analysis computes a metric M and uses it for hypothesis testing. Ricker et al. propose AEROBLADE (Ricker et al., 2024), a detector that uses the reconstruction error obtained from the pre-trained autoencoder of a latent diffusion model (LDM) for detection, where AI-generated images are found to have lower reconstruction errors than real images. He el al. propose RIGID (He et al., 2024), which computes the cosine similarity between the representations of a pair of the original and perturbed images subject to Gaussian noise, using the DINOV2 model (Oquab et al., 2023) for feature extraction. They find that real images tend to have higher cosine similarity to their perturbed versions than AI-generated images.